Assignment-2

Data Visualization and Analytics Project: Social Media Trends Analytics Dashboard

Project Link: https://github.com/akshattripathi1/Social-Media-Trends-Dashboard

Dataset: Custom Social Media Trends Dataset (2000 records, 12 columns)

Project Overview

This project presents a multi-page interactive dashboard developed using Python, Tkinter, Pandas, and Matplotlib to analyze social media trends and engagement metrics across different platforms.

The system transforms raw social media data into a visual analytics platform that helps users understand platform performance, content engagement patterns, and trending hashtags.

The dashboard is designed to provide insights into:

- Social media engagement trends across Twitter, Instagram, YouTube, Facebook, and TikTok
- Hashtag performance and trending analysis across 10 content categories
- Platform-wise reach comparison (Views, Likes, Comments, Shares)
- Content category engagement patterns (Entertainment, Sports, Technology, etc.)
- Metric correlation analysis (Posts vs Likes vs Views vs Engagement Rate)
- Platform × Category engagement heatmap for identifying content sweet spots

The interface follows a clean modern UI design with a midnight purple theme and color-coded platform indicators for better visual understanding.

Dataset Overview

The dataset contains 2000 records of social media trend data simulating realistic engagement patterns across multiple platforms and content categories.

It includes:

- Platform (Twitter, Instagram, YouTube, Facebook, TikTok)
- Hashtag name (60 unique hashtags)
- Content Category (Entertainment, Sports, Politics, Technology, Fashion, Gaming, Food, Travel, Education, Health)
- Country of origin (11 countries, India-biased)
- Date and Month

- Metrics:
  - o Posts count
  - o Likes
  - o Views
  - o Comments
  - o Shares
  - o Engagement Rate (%)

This dataset is used to analyze social media trends and visualize engagement patterns effectively.

Tech Stack

Python → Programming language

Tkinter → GUI framework for dashboard

Pandas → Data cleaning and analysis

Matplotlib → Data visualization (charts & graphs)

NumPy → Numerical computation and data processing

Program Code

```
"""
Social Media Trends Analytics Dashboard
DVA Assignment 2 | Akshat Tripathi | 04914202023 | IPU BCA 6th Sem
"""
import tkinter as tk
from tkinter import ttk
import pandas as pd
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
from matplotlib.figure import Figure
from matplotlib.gridspec import GridSpec
import os

# MIDNIGHT PURPLE THEME
```

```
BG="#13111C"; CARD="#1E1B2E"; CARD2="#252238"; FG="#F0E6FF"
ACCENT="#B983FF"; ACCENT2="#D4BBFF"; GOOD="#00E396"; WARN="#FF6178"
MID="#FEB019"; MUTED="#8A85A0"; BORDER="#362F50"
PLATFORM_COLORS={"Twitter":"#1DA1F2","Instagram":"#E4405F","YouTube":"#FF0000","Facebook":"#1877F2","TikTok":"#
CATEGORY_COLORS={"Entertainment":"#FF6B6B","Sports":"#4ECDC4","Politics":"#FFE66D","Technology":"#A29BFE","Fash
mpl.rcParams.update({"font.family":"sans-serif","font.sans-serif":["Segoe UI","DejaVu Sans","Arial"],"font.size
DPI=100; FILE_PATH="Social_Media_Trends_India.csv"
FONT_TABLE=("Segoe UI",10); FONT_THEAD=("Segoe UI",10,"bold")

def fmt_num(n):
    n=float(n)
    if n&gt;=1e6: return f"{n/1e6:.1f}M"
    if n&gt;=1e3: return f"{n/1e3:.1f}K"
    return str(int(n))

def load_data():
```

```
    df=pd.read_csv(FILE_PATH); df.columns=[c.strip().lower() for c in df.columns]; return df

def style_ax(ax,title="",xlabel="",ylabel=""):
    ax.set_facecolor(CARD)
    for sp in ax.spines.values(): sp.set_color(BORDER)
    ax.tick_params(axis="x",colors=FG); ax.tick_params(axis="y",colors=FG)
    if title: ax.set_title(title,color=ACCENT2,fontsize=12,fontweight="bold",pad=8)
    if xlabel: ax.set_xlabel(xlabel,color=MUTED)
    if ylabel: ax.set_ylabel(ylabel,color=MUTED)

def fix_yticks(ax,fs=10): ax.tick_params(axis="y",labelrotation=0,labelsize=fs,colors=FG)
def trunc(s,n=14): s=str(s); return s if len(s)&lt;=n else s[:n-1]+"..."

def hbar(ax,labels,values,colors,fs=10,bh=0.65,vlabels=True):
    bars=ax.barh(labels,values,color=colors,height=bh,edgecolor=BORDER,linewidth=0.5)
    if vlabels: ax.bar_label(bars,fmt="%.1f",color=FG,fontsize=fs-1,padding=3)
```

```
    fix_yticks(ax,fs); return bars

def embed(fig,parent):
    c=FigureCanvasTkAgg(fig,master=parent); c.draw(); c.get_tk_widget().pack(fill="both",expand=True)

def page_header(p,title,sub=""):
    hf=tk.Frame(p,bg=CARD,highlightbackground=ACCENT,highlightthickness=2)
    hf.pack(fill="x",padx=18,pady=(12,6))
    tk.Label(hf,text=title,bg=CARD,fg=ACCENT2,font=("Segoe UI",18,"bold")).pack(anchor="w",padx=16,pady=(10,2))
    if sub: tk.Label(hf,text=sub,bg=CARD,fg=MUTED,font=("Segoe UI",10)).pack(anchor="w",padx=16,pady=(0,8))

def cell_tc(cmap,nv):
    if nv is None or (isinstance(nv,float) and np.isnan(nv)): return FG
    rgba=cmap(float(nv)); lum=0.299*rgba[0]+0.587*rgba[1]+0.114*rgba[2]
    return "black" if lum&gt;0.50 else "white"
```

```
def scrollable_table(parent,hdrs,cwidths,rows_data,row_colors_fn):
    fr=tk.Frame(parent,bg=CARD,highlightbackground=ACCENT,highlightthickness=1)
    fr.pack(side="left",fill="both",expand=True,padx=(0,10))
    cv=tk.Canvas(fr,bg=CARD,highlightthickness=0)
    vs=ttk.Scrollbar(fr,orient="vertical",command=cv.yview)
    vs.pack(side="right",fill="y"); cv.pack(side="left",fill="both",expand=True)
    cv.configure(yscrollcommand=vs.set)
    inner=tk.Frame(cv,bg=CARD); win=cv.create_window((0,0),window=inner,anchor="nw")
    inner.bind("&lt;Configure&gt;",lambda e:cv.configure(scrollregion=cv.bbox("all")))
    cv.bind("&lt;Configure&gt;",lambda e:cv.itemconfig(win,width=max(e.width,inner.winfo_reqwidth())))
    cv.bind("&lt;MouseWheel&gt;",lambda e:cv.yview_scroll(int(-1*(e.delta/120)),"units"))
    for i,(h,w) in enumerate(zip(hdrs,cwidths)):
        tk.Label(inner,text=h,bg=BORDER,fg=ACCENT,font=FONT_THEAD,width=w,anchor="w").grid(row=0,column=i,padx=
    for ridx,vals in enumerate(rows_data,1):
        cbg=CARD2 if ridx%2==0 else CARD
        for cidx,(val,w) in enumerate(zip(vals,cwidths)):
```

```
            col=row_colors_fn(ridx,cidx,val)
            tk.Label(inner,text=val,bg=cbg,fg=col,font=FONT_TABLE,width=w,anchor="w").grid(row=ridx,column=cidx

def build_dashboard(root,df):
    # HEADER with accent border
    hdr=tk.Frame(root,bg=CARD,highlightbackground=ACCENT,highlightthickness=2)
    hdr.pack(fill="x",padx=16,pady=(10,4))
    lh=tk.Frame(hdr,bg=CARD); lh.pack(side="left",padx=14,pady=10)
    tk.Label(lh,text="Social Media Trends Analytics Dashboard",bg=CARD,fg=FG,font=("Segoe UI",22,"bold")).pack(
    tk.Label(lh,bg=CARD,fg=MUTED,font=("Segoe UI",10),text=f"{len(df)} Records  |  {df['platform'].nunique()} P
    rh=tk.Frame(hdr,bg=CARD); rh.pack(side="right",padx=14,pady=10)
    tk.Label(rh,text="DVA Assignment 2",bg=CARD,fg=ACCENT,font=("Segoe UI",12,"bold")).pack(anchor="e")
    tk.Label(rh,text="Akshat Tripathi | 04914202023",bg=CARD,fg=ACCENT2,font=("Segoe UI",10,"bold")).pack(ancho
    tk.Label(rh,text="IPU BCA 6th Semester",bg=CARD,fg=MUTED,font=("Segoe UI",9)).pack(anchor="e")

    # NOTEBOOK
```

```
    st=ttk.Style(); st.theme_use("default")
    st.configure("TNotebook",background=BG,borderwidth=0)
    st.configure("TNotebook.Tab",background=CARD,foreground=FG,padding=(28,11),font=("Segoe UI",11,"bold"))
    st.map("TNotebook.Tab",background=[("selected","#B983FF")],foreground=[("selected","#13111C")])
    nb=ttk.Notebook(root,style="TNotebook"); nb.pack(fill="both",expand=True,padx=12,pady=8)
    page1(nb,df); page2(nb,df); page3(nb,df); page4(nb,df); page5(nb,df)

def page1(nb,df):
    p=tk.Frame(nb,bg=BG); nb.add(p,text="  Overview  ")
    body=tk.Frame(p,bg=BG); body.pack(fill="both",expand=True)
    left=tk.Frame(body,bg=BG,width=300); left.pack(side="left",fill="y",padx=(8,6),pady=8); left.pack_propagate
    kpis=[
        ("Total Posts",fmt_num(df["posts"].sum()),"All platforms combined",ACCENT),
        ("Total Views",fmt_num(df["views"].sum()),"Combined reach",ACCENT2),
        ("Total Likes",fmt_num(df["likes"].sum()),"User reactions",GOOD),
        ("Avg Engagement",f"{df['engagement_rate'].mean():.2f}%","(Likes+Cmts+Shares)/Views",MID),
```

```
    ("Top Platform",df.groupby("platform")["views"].sum().idxmax(),"By total views",PLATFORM_COLORS.get(df.
    ("Trending Category",df.groupby("category")["engagement_rate"].mean().idxmax(),"Highest avg engagement"
]
for title,val,sub,c in kpis:
    card=tk.Frame(left,bg=CARD,padx=12,pady=8,highlightbackground=BORDER,highlightthickness=1); card.pack(f
    tk.Label(card,text=title,bg=CARD,fg=MUTED,font=("Segoe UI",10)).pack(anchor="w")
    tk.Label(card,text=val,bg=CARD,fg=c,font=("Segoe UI",15,"bold")).pack(anchor="w")
    if sub: tk.Label(card,text=sub,bg=CARD,fg=BORDER,font=("Segoe UI",8)).pack(anchor="w")
right=tk.Frame(body,bg=BG); right.pack(side="left",fill="both",expand=True,padx=(8,18),pady=8)
fig=Figure(figsize=(10.5,5.5),dpi=DPI); fig.patch.set_facecolor(BG)
gs=GridSpec(2,2,figure=fig,hspace=0.50,wspace=0.40)
ax1=fig.add_subplot(gs[0,0])
pv=df.groupby("platform")["views"].sum().sort_values()
hbar(ax1,pv.index,pv.values/1e6,[PLATFORM_COLORS.get(p,ACCENT) for p in pv.index],fs=9,bh=0.55,vlabels=Fals
style_ax(ax1,"Platform Reach (Million Views)")
ax2=fig.add_subplot(gs[1,0])
```

```
cat_cnt=df["category"].value_counts(); clrs=[CATEGORY_COLORS.get(c,ACCENT) for c in cat_cnt.index]
wedges,_,autotexts=ax2.pie(cat_cnt.values,colors=clrs,autopct=lambda p:f"{p:.0f}%" if p&gt;=5 else "",pctdi
for at,wdg in zip(autotexts,wedges):
    fc=wdg.get_facecolor(); lum=0.299*fc[0]+0.587*fc[1]+0.114*fc[2]
    at.set_fontsize(8); at.set_fontweight("bold"); at.set_color("black" if lum&gt;0.5 else "white")
ax2.legend(wedges,[f"{c}({n})" for c,n in cat_cnt.items()],loc="upper center",bbox_to_anchor=(0.5,0.0),ncol
ax2.set_title("Category Distribution",color=ACCENT2,fontsize=11,fontweight="bold",pad=8)
ax3=fig.add_subplot(gs[0,1])
ht=df.groupby("hashtag")["views"].sum().nlargest(5).sort_values()
hbar(ax3,ht.index,ht.values/1e6,[ACCENT]*5,fs=9,bh=0.6)
style_ax(ax3,"Top 5 Hashtags (M Views)")
ax4=fig.add_subplot(gs[1,1])
pe=df.groupby("platform")["engagement_rate"].mean().sort_values()
hbar(ax4,pe.index,pe.values,[PLATFORM_COLORS.get(p,ACCENT) for p in pe.index],fs=9,bh=0.55)
ax4.axvline(3,color=MID,lw=1.3,ls="--",label="Avg 3%"); ax4.legend(fontsize=8,labelcolor=FG,facecolor=CARD,
style_ax(ax4,"Avg Engagement Rate (%)")
```

```
    fig.tight_layout(pad=1.5); embed(fig,right)

def page2(nb,df):
    p=tk.Frame(nb,bg=BG); nb.add(p,text="  Platform Analysis  ")
    page_header(p,"Platform-wise Social Media Performance","Detailed metrics comparison across all 5 platforms"
    body=tk.Frame(p,bg=BG); body.pack(fill="both",expand=True,padx=16,pady=6)
    plat_df=df.groupby("platform").agg(posts=("posts","sum"),likes=("likes","sum"),views=("views","sum"),commen
    hdrs=["PLATFORM","POSTS","LIKES","VIEWS","COMMENTS","SHARES","ENG%"]; cw=[14,11,11,12,11,11,9]
    rows=[[r["platform"],fmt_num(r["posts"]),fmt_num(r["likes"]),fmt_num(r["views"]),fmt_num(r["comments"]),fmt
    def colorfn(ri,ci,val): return PLATFORM_COLORS.get(str(val),FG) if ci==0 else FG
    scrollable_table(body,hdrs,cw,rows,colorfn)
    ch=tk.Frame(body,bg=BG); ch.pack(side="left",fill="both",expand=True)
    fig=Figure(figsize=(8.5,6.0),dpi=DPI); fig.patch.set_facecolor(BG)
    gs=GridSpec(2,2,figure=fig,hspace=0.65,wspace=0.46)
    plats=plat_df["platform"]; clrs=[PLATFORM_COLORS.get(p,ACCENT) for p in plats]
    for i,(col,title) in enumerate([("likes","Likes (M)"),("shares","Shares (M)"),("comments","Comments (M)"),(
```

```
        ax=fig.add_subplot(gs[i//2,i%2])
        vals=plat_df[col]/1e6 if col!="avg_eng" else plat_df[col]
        ax.bar(plats,vals,color=clrs,edgecolor=BORDER,linewidth=0.5); style_ax(ax,title); ax.tick_params(axis="
    fig.subplots_adjust(left=0.12,right=0.97,top=0.94,bottom=0.12); embed(fig,ch)

def page3(nb,df):
    p=tk.Frame(nb,bg=BG); nb.add(p,text="  Category Trends  ")
    page_header(p,"Content Category Performance Analysis","Engagement and reach metrics by content category")
    fig=Figure(figsize=(12,6.5),dpi=DPI); fig.patch.set_facecolor(BG)
    gs=GridSpec(2,2,figure=fig,hspace=0.50,wspace=0.35)
    cat_df=df.groupby("category").agg(views=("views","sum"),eng=("engagement_rate","mean")).sort_values("views"
    ax1=fig.add_subplot(gs[0,0])
    ax1.barh(cat_df.index,cat_df["views"]/1e6,color=[CATEGORY_COLORS.get(c,ACCENT) for c in cat_df.index],heigh
    style_ax(ax1,"Views by Category (M)"); fix_yticks(ax1,9)
    ax2=fig.add_subplot(gs[0,1])
    ce=cat_df.sort_values("eng"); ax2.barh(ce.index,ce["eng"],color=[CATEGORY_COLORS.get(c,ACCENT) for c in ce.
```

```
    ax2.axvline(3,color=MID,lw=1.3,ls="--"); style_ax(ax2,"Avg Engagement Rate (%)"); fix_yticks(ax2,9)
    ax3=fig.add_subplot(gs[1,0])
    ht=df.groupby("hashtag")["engagement_rate"].mean().nlargest(10).sort_values()
    ax3.barh(ht.index,ht.values,color=[ACCENT2]*10,height=0.6,edgecolor=BORDER,linewidth=0.5); style_ax(ax3,"To
    ax4=fig.add_subplot(gs[1,1])
    cat_posts=df["category"].value_counts(); clrs=[CATEGORY_COLORS.get(c,ACCENT) for c in cat_posts.index]
    wedges,_,ats=ax4.pie(cat_posts.values,colors=clrs,autopct=lambda p:f"{p:.0f}%" if p&gt;=5 else "",pctdistan
    for at,wdg in zip(ats,wedges):
        fc=wdg.get_facecolor(); lum=0.299*fc[0]+0.587*fc[1]+0.114*fc[2]; at.set_fontsize(8); at.set_fontweight(
    ax4.set_title("Posts per Category",color=ACCENT2,fontsize=11,fontweight="bold",pad=8)
    ax4.legend(wedges,cat_posts.index,loc="upper center",bbox_to_anchor=(0.5,0.0),ncol=2,fontsize=7,labelcolor=
    fig.subplots_adjust(left=0.18,right=0.96,top=0.92,bottom=0.08); embed(fig,p)

def page4(nb,df):
    p=tk.Frame(nb,bg=BG); nb.add(p,text="  Hashtag Rankings  ")
    page_header(p,"Hashtag Performance Rankings","Top and bottom performing hashtags by total views")
```

```
outer=tk.Frame(p,bg=BG); outer.pack(fill="both",expand=True,padx=16,pady=6)
ht_df=df.groupby("hashtag").agg(category=("category","first"),views=("views","sum"),likes=("likes","sum"),e
hdrs=["#","HASHTAG","CATEGORY","VIEWS","LIKES","ENG%","POSTS"]; cw=[4,16,14,11,11,9,11]
rows=[[i+1,r["hashtag"],r["category"],fmt_num(r["views"]),fmt_num(r["likes"]),f"{r['eng']:.2f}%",fmt_num(r[
def colorfn(ri,ci,val):
    if ci==0: return MUTED
    if ci==2: return CATEGORY_COLORS.get(str(val),FG)
    return FG
scrollable_table(outer,hdrs,cw,rows,colorfn)
frr=tk.Frame(outer,bg=CARD,highlightbackground=ACCENT,highlightthickness=1); frr.pack(side="left",fill="bot
fig=Figure(figsize=(6.8,6.0),dpi=DPI); fig.patch.set_facecolor(CARD)
ax1=fig.add_subplot(211); ax1.set_facecolor(CARD)
t10=ht_df.head(10).sort_values("views"); hbar(ax1,t10["hashtag"],t10["views"]/1e6,[WARN]*10,fs=9,bh=0.60,vl
ax1.set_title("[HIGH] Top 10 Most Viewed",color=ACCENT2,fontsize=10,fontweight="bold",pad=6); fix_yticks(ax
ax2=fig.add_subplot(212); ax2.set_facecolor(CARD)
b10=ht_df.tail(10).sort_values("views",ascending=False); hbar(ax2,b10["hashtag"],b10["views"]/1e6,[GOOD]*10
```

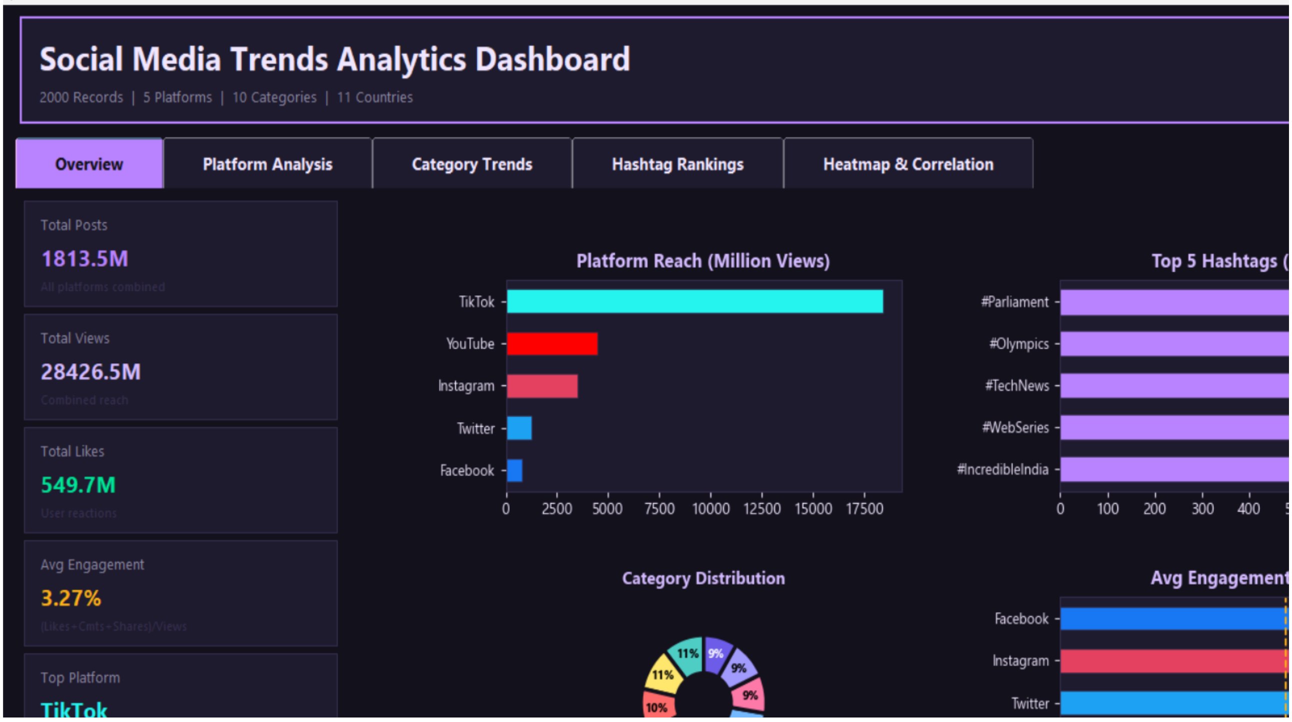
```
    ax2.set_title("[LOW] Bottom 10 Least Viewed",color=ACCENT2,fontsize=10,fontweight="bold",pad=6); fix_yticks
    fig.subplots_adjust(left=0.32,right=0.96,top=0.92,bottom=0.08,hspace=0.60); embed(fig,frr)

def page5(nb,df):
    p=tk.Frame(nb,bg=BG); nb.add(p,text="  Heatmap &amp; Correlation  ")
    page_header(p,"Metric Correlation &amp; Platform x Category Heatmap","Left: inter-metric correlation  |  Ri
    fig=Figure(figsize=(12,5.8),dpi=DPI); fig.subplots_adjust(left=0.10,right=0.96,top=0.88,bottom=0.18,wspace=
    cols=["posts","likes","views","comments","shares","engagement_rate"]; lbls=["Posts","Likes","Views","Commen
    corr=df[cols].corr()
    ax1=fig.add_subplot(121)
    im1=ax1.imshow(corr.values,cmap="PuOr",vmin=-1,vmax=1,aspect="auto")
    ax1.set_xticks(range(len(lbls))); ax1.set_xticklabels(lbls,rotation=35,ha="right",fontsize=10,color=FG)
    ax1.set_yticks(range(len(lbls))); ax1.set_yticklabels(lbls,fontsize=10,color=FG)
    cmap1=plt.get_cmap("PuOr")
    for i in range(len(lbls)):
        for j in range(len(lbls)):
```

```
        v=corr.values[i,j]; tc=cell_tc(cmap1,(v+1)/2)
        ax1.text(j,i,f"{v:.2f}",ha="center",va="center",fontsize=10,fontweight="bold",color=tc)
cb1=fig.colorbar(im1,ax=ax1,shrink=0.80); cb1.ax.yaxis.set_tick_params(color=FG,labelsize=10); plt.setp(cb1
style_ax(ax1,"Metric Correlation Matrix")
pivot=df.pivot_table(index="platform",columns="category",values="engagement_rate",aggfunc="mean")
norm=(pivot-pivot.min())/(pivot.max()-pivot.min())
ax2=fig.add_subplot(122)
cmap2=plt.get_cmap("magma")
im2=ax2.imshow(norm.values,cmap="magma",aspect="auto",vmin=0,vmax=1)
ax2.set_xticks(range(len(pivot.columns))); ax2.set_xticklabels([trunc(c,9) for c in pivot.columns],fontsize
ax2.set_yticks(range(len(pivot.index))); ax2.set_yticklabels(pivot.index,fontsize=10,color=FG)
for i in range(len(pivot.index)):
    for j in range(len(pivot.columns)):
        rv=pivot.values[i,j]; nv=norm.values[i,j]
        txt="nan" if (isinstance(rv,float) and np.isnan(rv)) else f"{rv:.1f}"
        tc=cell_tc(cmap2,None if (isinstance(nv,float) and np.isnan(nv)) else nv)
```

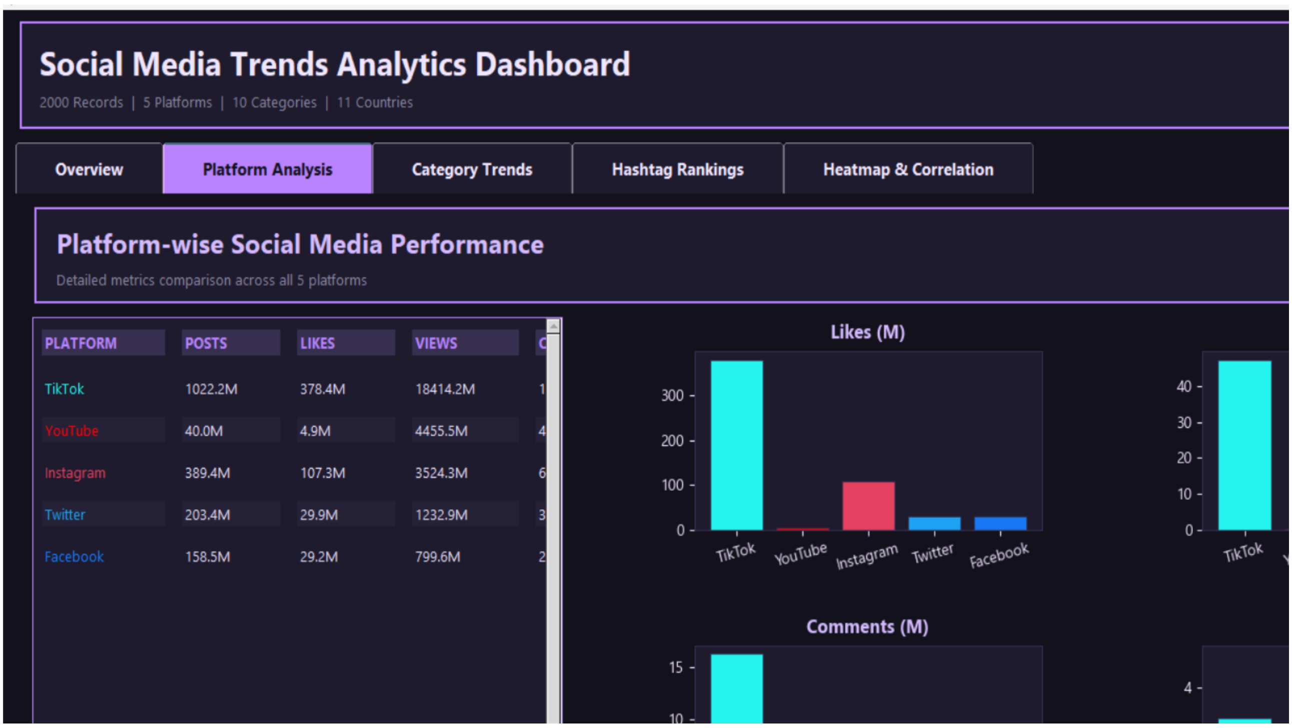
```
            ax2.text(j,i,txt,ha="center",va="center",fontsize=9,color=tc)
    cb2=fig.colorbar(im2,ax=ax2,shrink=0.80); cb2.ax.yaxis.set_tick_params(color=FG,labelsize=10); plt.setp(cb2
    style_ax(ax2,"Platform x Category Engagement")
    embed(fig,p)

def main():
    root=tk.Tk(); root.title("Social Media Trends Dashboard - Akshat Tripathi | 04914202023")
    root.configure(bg=BG); root.state("zoomed")
    df=load_data(); build_dashboard(root,df); root.mainloop()

if __name__=="__main__":
    main()
```

Output

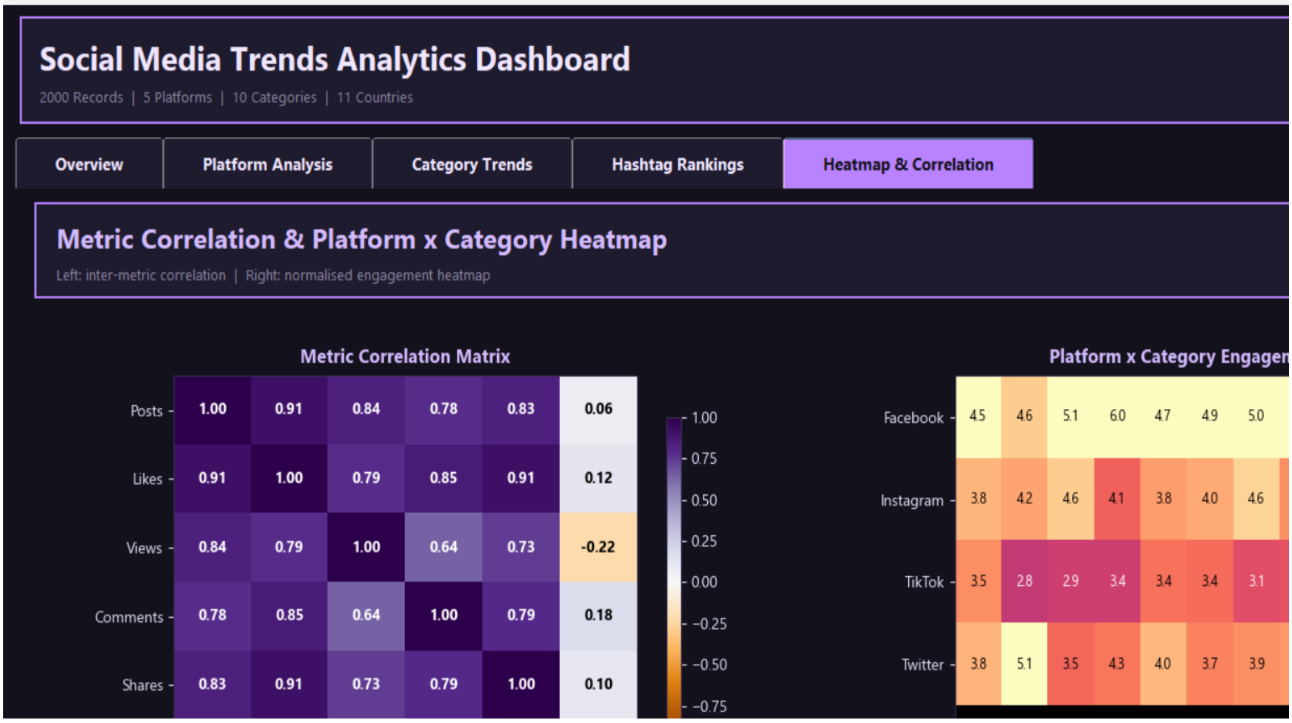
Page 1- Overview



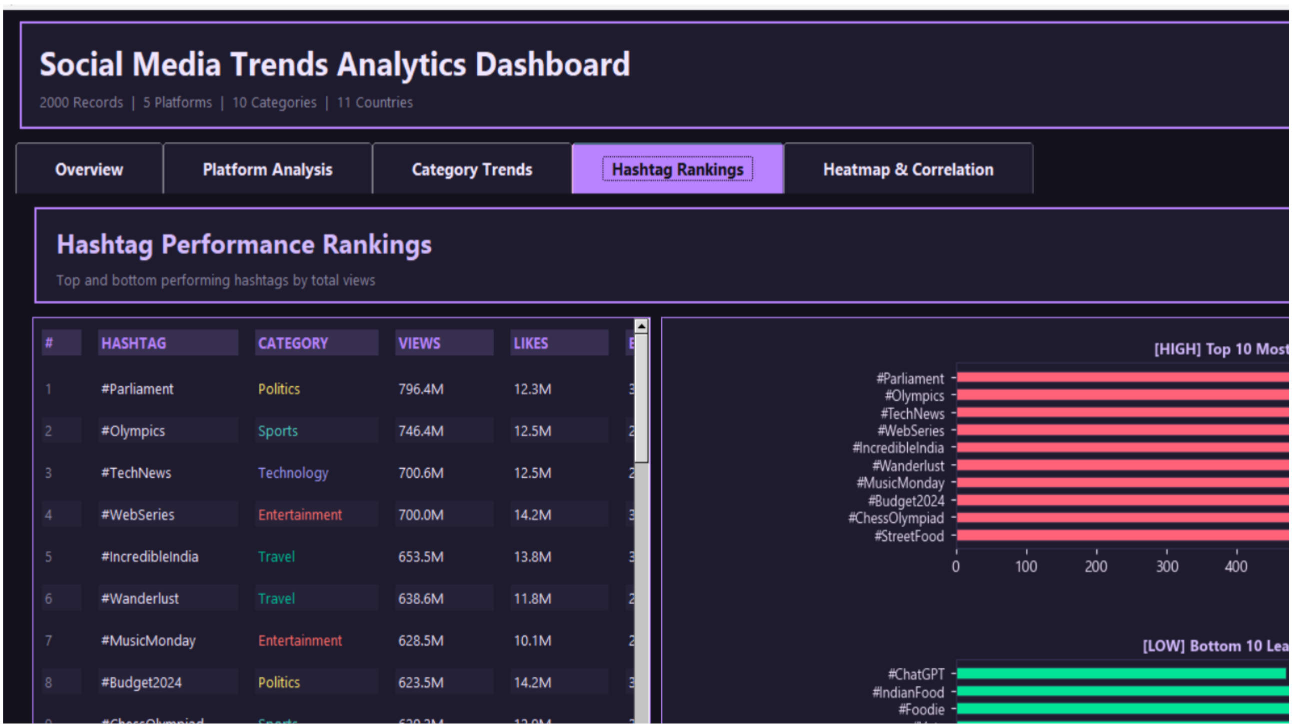
Page 2- Platform Analysis



Page 3- Category Trends



Page 4- Hashtag Rankings



Page 5- Heatmap & Correlations